

9.2.16

EE25BTECH11054 - S.Harsha Vardhan

Question 9.2.16 Find the area of the region bounded by $y = \sqrt{x}$ and $y = x$.

Solution: The given curves are $y^2 = x$ (for $y \geq 0$) and $y = x$. The matrix form of the parabola $y^2 - x = 0$ is $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$, with parameters:

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -1/2 \\ 0 \end{pmatrix}, \quad f = 0. \quad (1)$$

The parametric form of the line $y = x$ is $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$, with parameters:

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (2)$$

The intersection parameter κ :

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (3)$$

where $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$. With $\mathbf{h} = \mathbf{0}$, we have $g(\mathbf{h}) = 0$ and $\mathbf{V} \mathbf{h} = \mathbf{0}$. The terms for the formula are:

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \quad (4)$$

$$\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) = \mathbf{m}^\top \mathbf{u} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} -1/2 \\ 0 \end{pmatrix} = -1/2 \quad (5)$$

Substituting into equation (3):

$$\kappa_i = \frac{-(-1/2) \pm \sqrt{(-1/2)^2 - 0}}{1} = \frac{1}{2} \pm \frac{1}{2} \quad (6)$$

The values for κ are:

$$\kappa_1 = 1, \quad \kappa_2 = 0 \quad (7)$$

The points of intersection are found by substituting κ_i into the line equation:

$$\mathbf{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (8)$$

$$\mathbf{x}_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (9)$$

The area is given by the integral:

$$A = \int_0^1 (\sqrt{x} - x) dx \quad (10)$$

$$= \left[\frac{2}{3} x^{3/2} - \frac{1}{2} x^2 \right]_0^1 \quad (11)$$

$$= \left(\frac{2}{3} - \frac{1}{2} \right) - (0) \quad (12)$$

$$= \frac{1}{6} \quad (13)$$

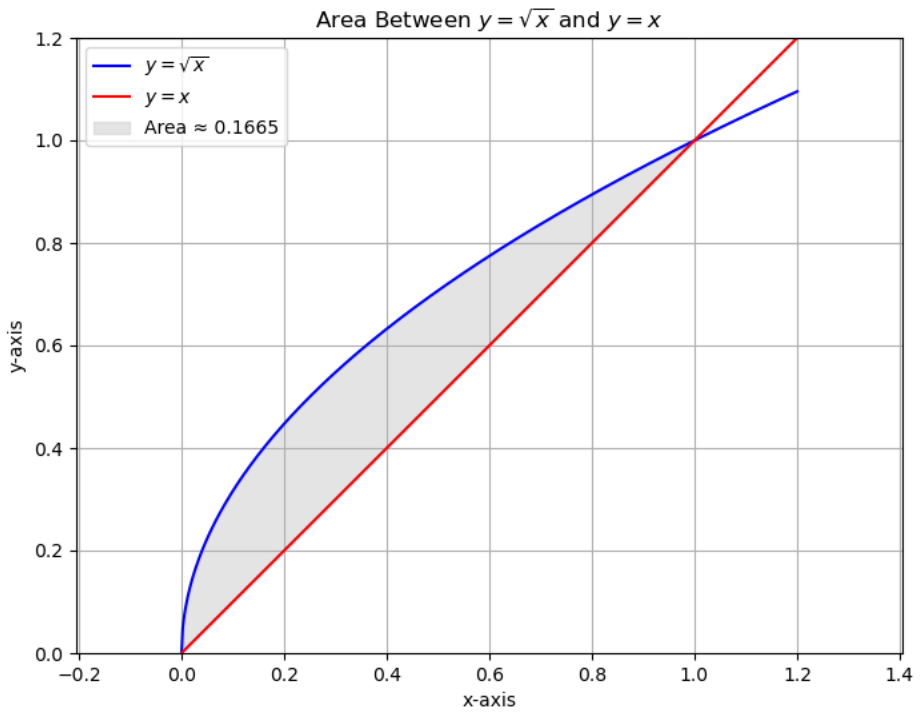


Fig. 0.1